

Unit 11, Lesson 1: Constructing a Circumscribed Circle of a Triangle



Benchmark

MA.912.GR.5.3 Construct the inscribed and circumscribed circles of a triangle.



Learning Targets

- ☐ I can construct the circumscribed circle of a triangle.
- ☐ I can locate the circumcenter of a triangle.



11.1.1 Warm-Up: Lake Okeechobee and Deep Lake

Lake Okeechobee means "big water" in the language of the Seminole tribe. Lake Okeechobee has a surface area of 730 square miles. Deep Lake in Big Cypress National Preserve is a naturally occurring sinkhole that was formed roughly 6,000 years ago. Deep Lake has a surface area of 0.002 square miles.

1. How many Deep Lakes are needed to cover the same surface area as Lake Okeechobee?

Lake Okeechobee averages 9 feet in depth, whereas Deep Lake averages 95 feet in depth. Based on the volume of the two lakes it would take approximately 30,850 Deep Lakes to fill up Lake Okeechobee.

2. Explain why it takes more Deep Lakes to cover the surface area of Lake Okeechobee than to fill up Lake Okeechobee.



11.1.2 Exploration: Circumscribing A Triangle With A Circle

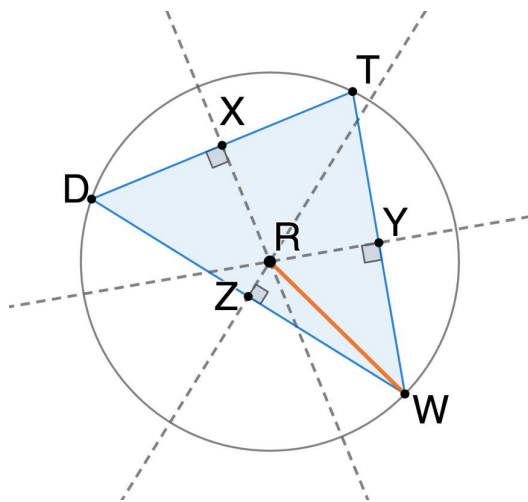


An **inscribed polygon in a circle** is a polygon which has all of its vertices on a circle.

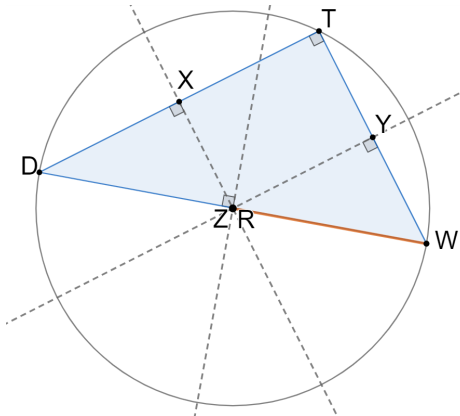
$\triangle TWD$ is inscribed in Circle R , where $\overline{RW}$ is a radius. Lines $\overline{RX}$, $\overline{RY}$, and $\overline{RZ}$ are perpendicular bisectors of $\overline{TD}$, $\overline{TW}$, and $\overline{DW}$, respectively.

Each diagram shows a different type of triangle inscribed in Circle R .

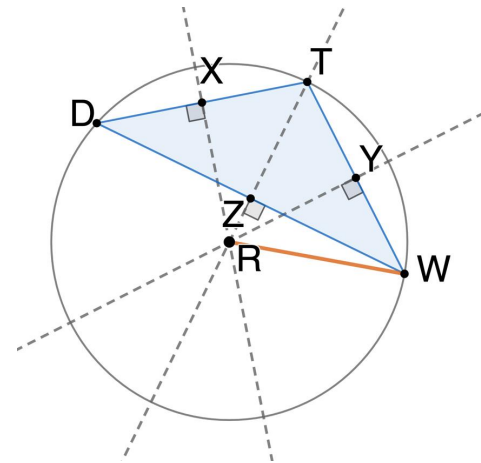
Acute



Right



Obtuse



- Determine similarities and differences among all three inscribed triangles.

Similarities	Differences

- Point R is the center of the circle that is circumscribed about $\triangle TWD$. Make a conjecture about point R as it relates to the perpendicular bisectors of the sides of $\triangle TWD$.

3. Ask two other pairs of partners for the conjectures they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

My Summary of Their Conjecture	Initials

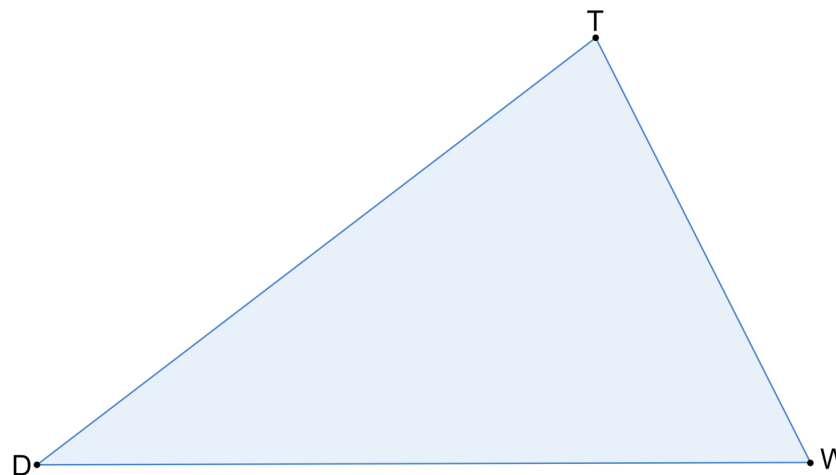
4. Review your conjecture with your partner and if necessary, make any changes.



A **circumscribed circle** is the smallest circle that includes a plane figure. If the figure is a polygon, then the circle must contain all of the vertices of the polygon. Not every polygon has a circumscribed circle, but all triangles and all regular polygons have circumscribed circles.

The **point of concurrency** is the point where three or more lines intersect.

5. Copy $\triangle DTW$ onto patty paper. Construct the perpendicular bisectors of each side. Name the point of concurrency of $\triangle DTW$ point R .



6. Discuss with your partner the relationship point R has with each of the vertices.

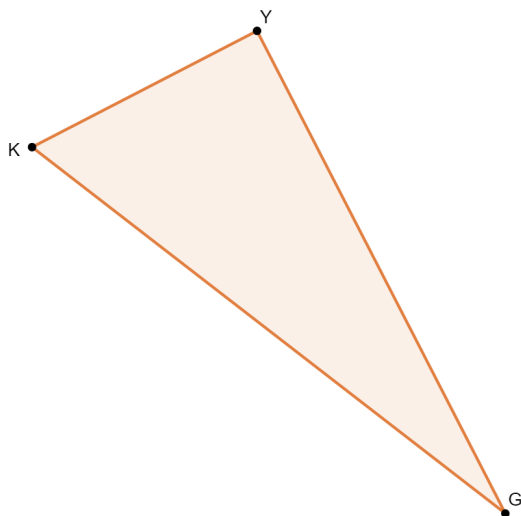
The point of concurrency of the perpendicular bisectors of a triangle (the center of a triangle's circumscribed circle) is the **circumcenter**.

7. Use the definition of a circumcenter to explain how the relationship point R has with each of the vertices ensures that a circle can be drawn to circumscribe the triangle.

8. Use a compass centered at point R to draw the circle that will pass through all three vertices.

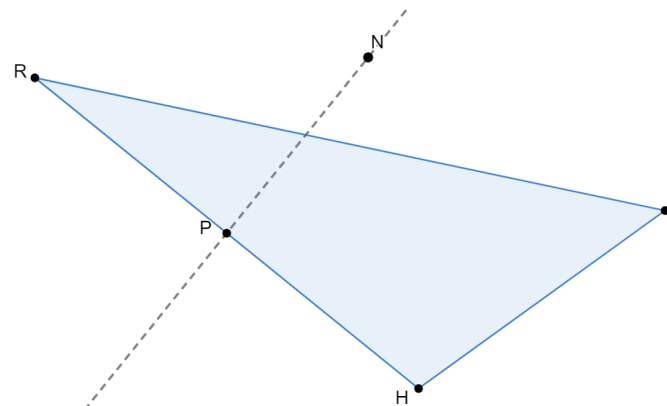
9. Review your conjecture. Does your conjecture hold true for newly constructed circumscribed Circle R ?

10. Copy $\triangle KYG$ onto another piece of paper. Then, use a compass and a straight edge to construct the circle that circumscribes $\triangle KYG$.



11. Describe how the circumscribed circle can be used to determine the type of triangle that $\triangle KYG$ is. Name a property and theorem from Unit 10 that can be used to verify this.

12. The dotted line is the perpendicular bisector of side $\overline{RH}$. The distance between points R and N is 4.35 units. If point N is the circumcenter, what is the distance between points H and N ? Explain or show your reasoning.



13. Emilio tells his partner, Talia, that in order to show that point N is the circumcenter, he only needs to construct one more perpendicular bisector. Talia challenged him to prove it. Describe how Emilio could justify that point N is the circumcenter by drawing only one more perpendicular bisector.



11.1.3 Wrap-Up: Ticket Out the Door

1. Describe how to construct Circle H , the circumscribed circle of $\triangle CRW$.

